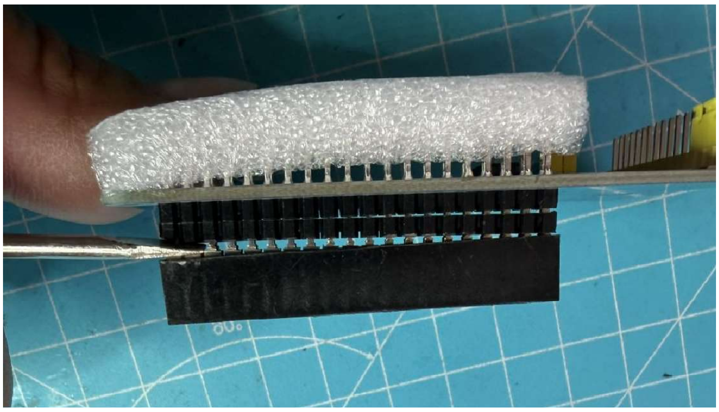
Assembly Instructions – AMS SpoolEdit

1. Cut the four threads for the bottom plate mounting into the housing base using an M3 tap.

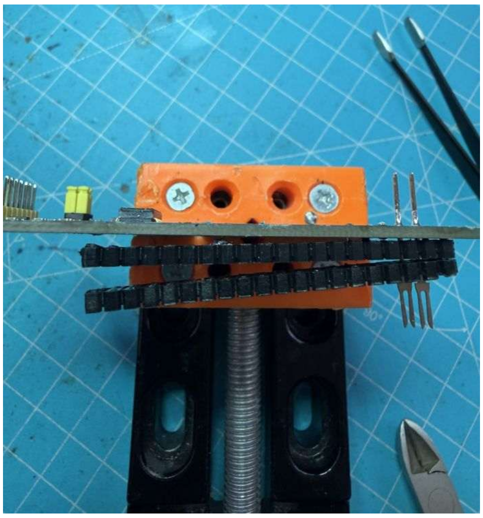
2. Prepare the NFC reader for installation. To reduce the required space inside the enclosure, the pin header needs to be modified.



Using a small screwdriver, as shown in the image, carefully remove the plastic housing of the pin header from the pins.

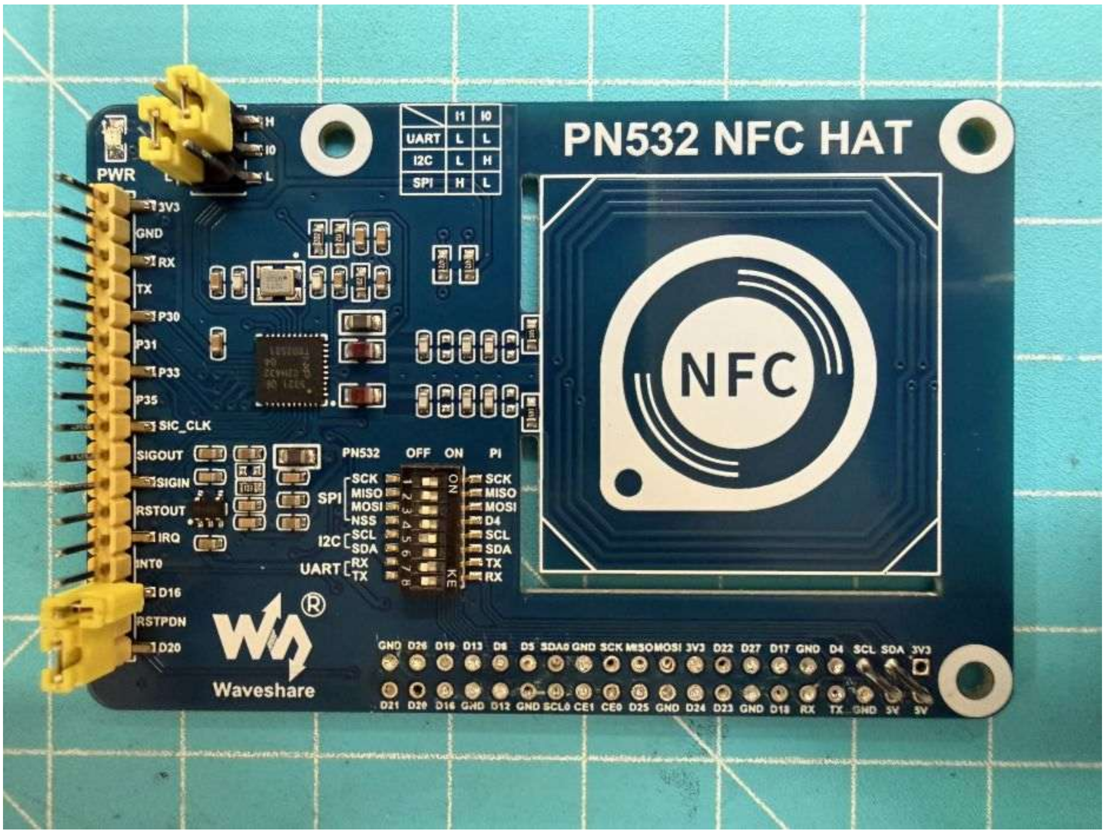
Once the housing has been removed, the unused pins can be desoldered. This is easiest if the reader is secured as shown in the image.

Using a soldering iron, heat the pins one at a time from the back side until the solder melts and the pins can be pulled out from the opposite side. Apply sufficient heat to each pin before attempting to remove it. The pins labeled GND require slightly more heat to remove. The two pins labeled SCL and SDA must remain in place and must not be desoldered.

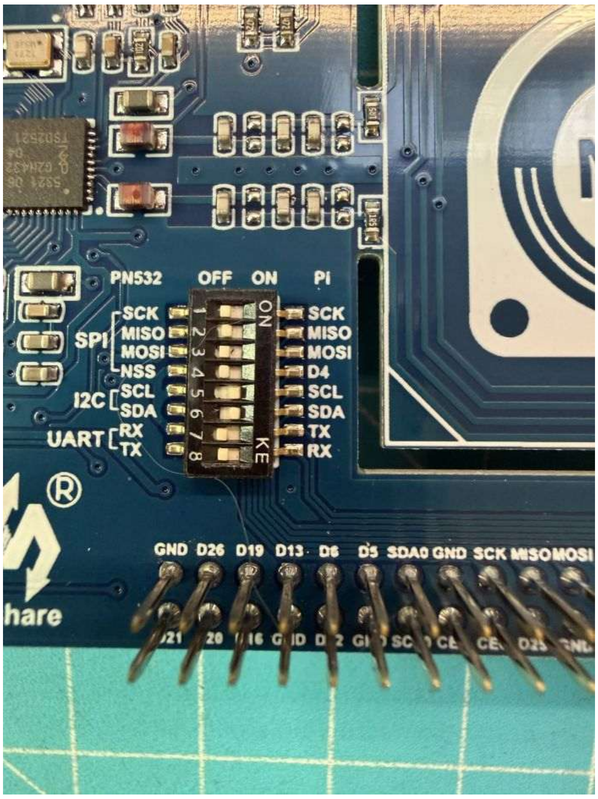


Next, use side cutters to cut these two pins as shown in the image, allowing the two plastic strips to be removed completely. On the side where the plastic strips were just removed, trim the remaining ends of the two pins flush with the PCB.

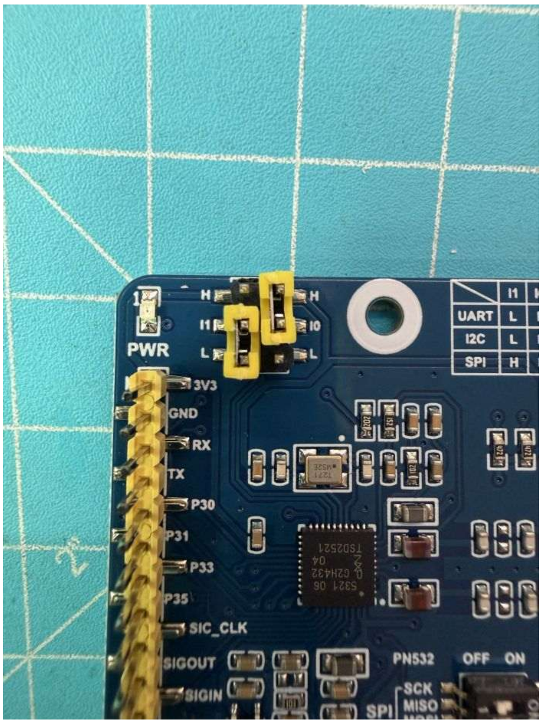
Once completed, the reader should look as shown below.

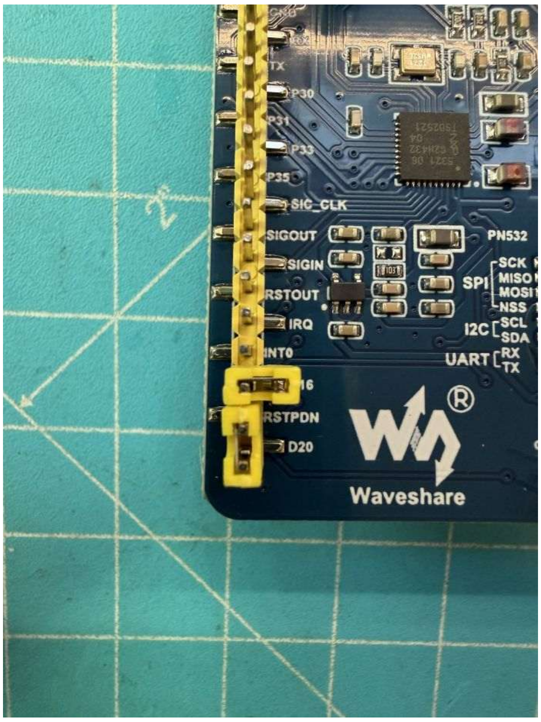


3. The reader now only needs to be configured correctly. Set the DIP switches as shown in the image: I2C SCL (5) and SDA (6) to ON, and all remaining switches to OFF.
4.



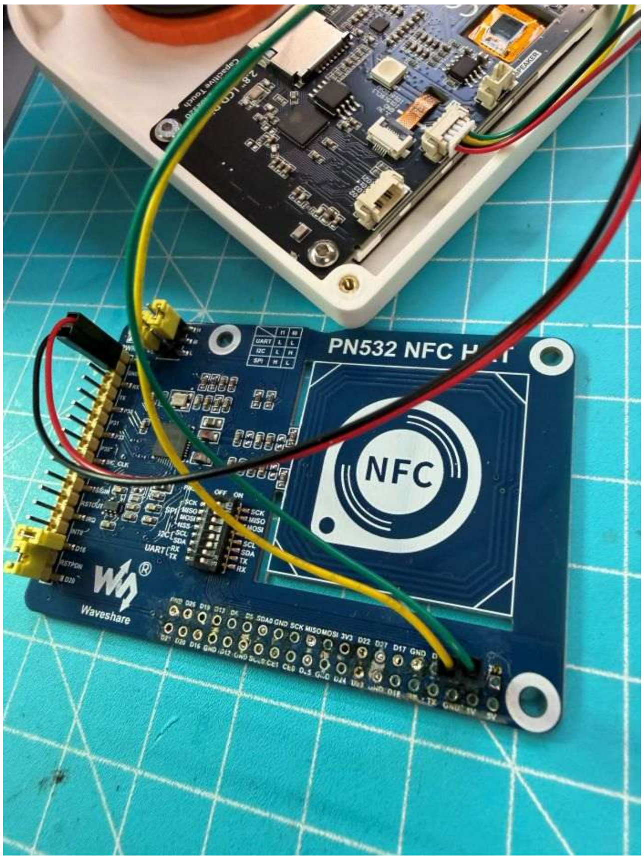
Move the jumpers in the upper-left corner to the left position, as shown in the image. Then install the pin header in the lower-left corner as shown in the image on the right.





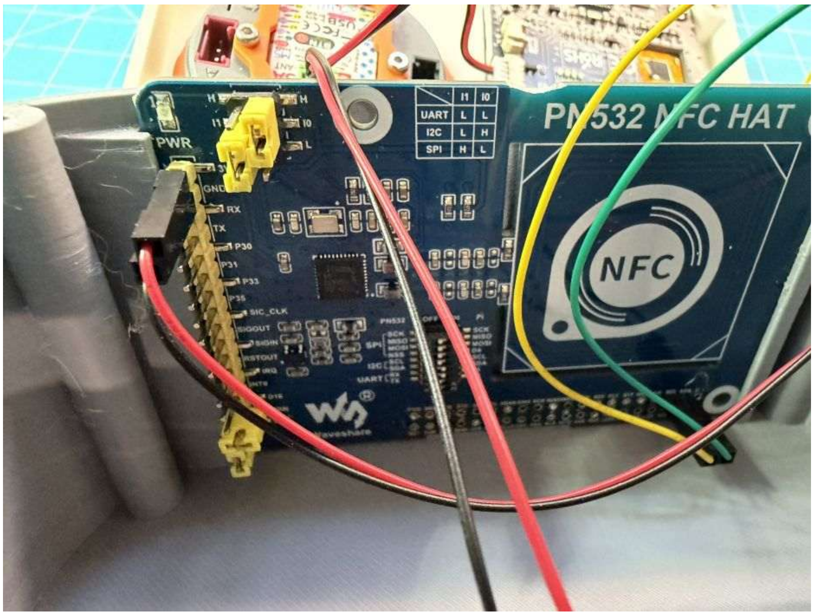
The reader is now ready for installation.

5. Connect the reader as shown in the following image.



RED =========→ 3,3 Volt
BLACK=== ===→ GND
YELLOW =====→ SCL
GREEN =====→ SDA

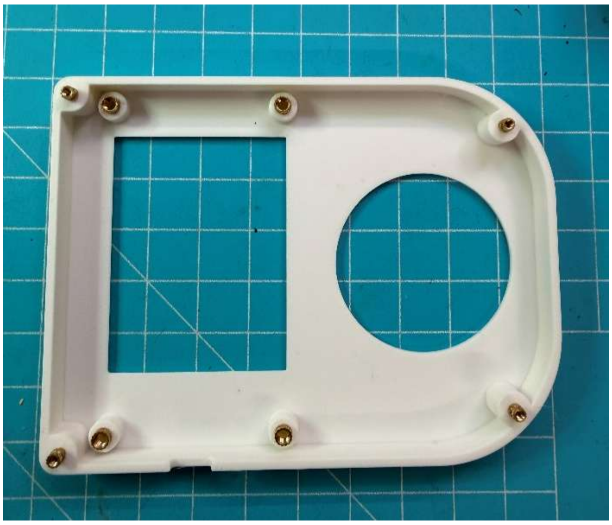
On the display, the reader will later be plugged in as shown in the image. However, it can already be installed in the enclosure.



6. Next, prepare the front side of the reader. To do this, the heat-set threaded inserts must be installed at the designated positions.

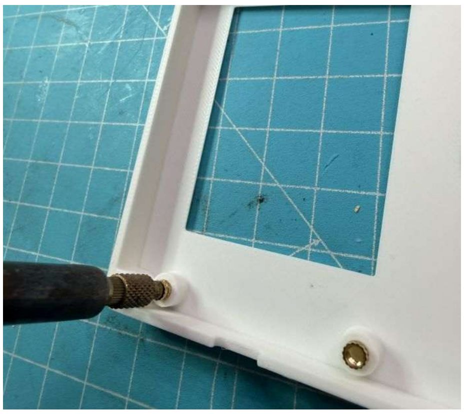
   The M3 inserts are used for mounting the display.
   The M2 threaded inserts are intended for joining the two housing halves.

   To position them, use a longer screw and press the cylindrical end of the insert into the designated hole. The remaining installation is then completed using a soldering iron.

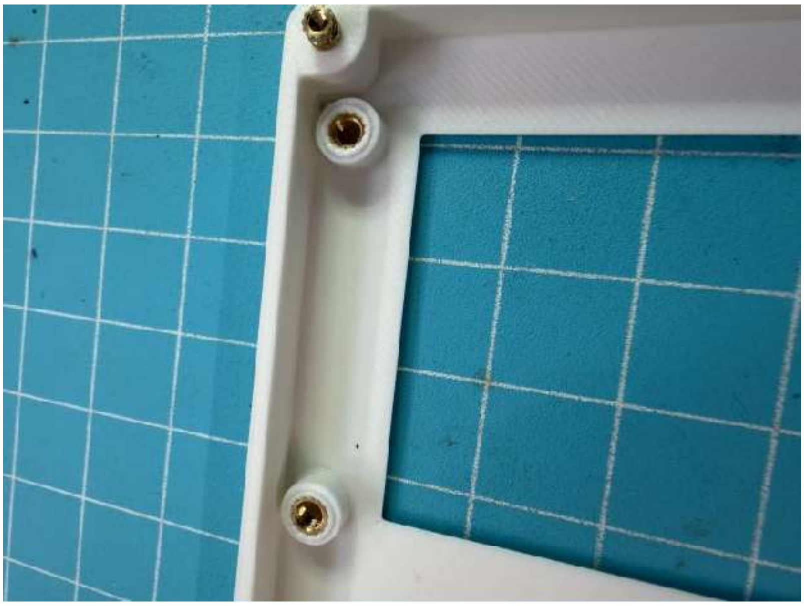



Once all inserts are positioned, it should look like this.

When heat-inserting the threaded inserts, make sure they are seated straight and flush. They must not protrude under any circumstances; if in doubt, it is better to set them slightly deeper.



Carefully trim off the small bead of material that is pushed out during the heat-insertion process using a utility knife once the material has cooled, so that it is flush.



Carefully remove any material that may have entered the threads using an appropriately sized tap.

Once completed, it should look like this.

7. Install the display and dial into the enclosure lid.
   First, secure the display using four M3 × 4 mm screws.
   Make sure the USB port is aligned with the cutout.

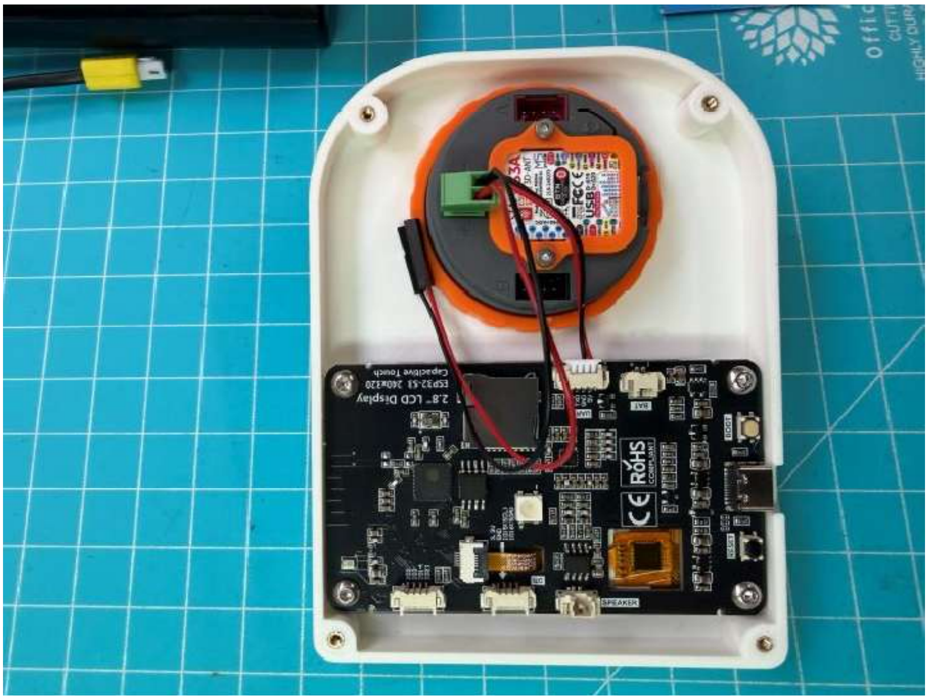
Next, install the dial. Make sure that the arrow on the front of the dial is pointing upwards.

Now connect the display and the dial as shown in the image.

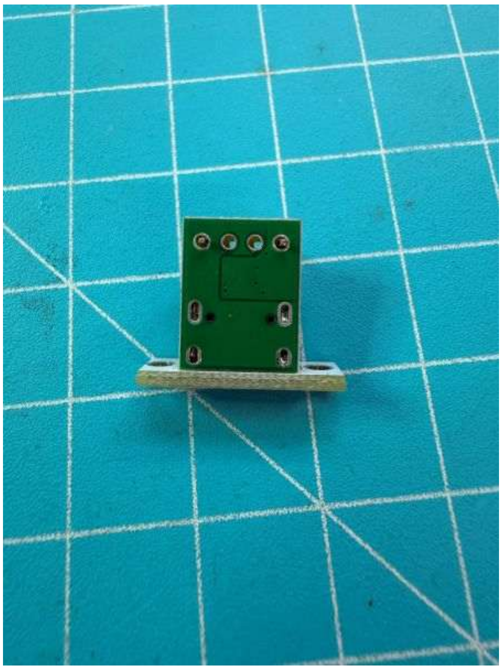
First, remove the yellow and green wires from the 4-pin connector by lifting the small retaining tabs, as they are not needed.
Next, cut the remaining cable so that the short section from the dial to the display is just long enough, with a little extra slack.

Strip approximately 3 mm of insulation from all four cut wire ends. Then twist the two red wire ends together and tin them with a soldering iron. Repeat the same process with the two black wire ends.

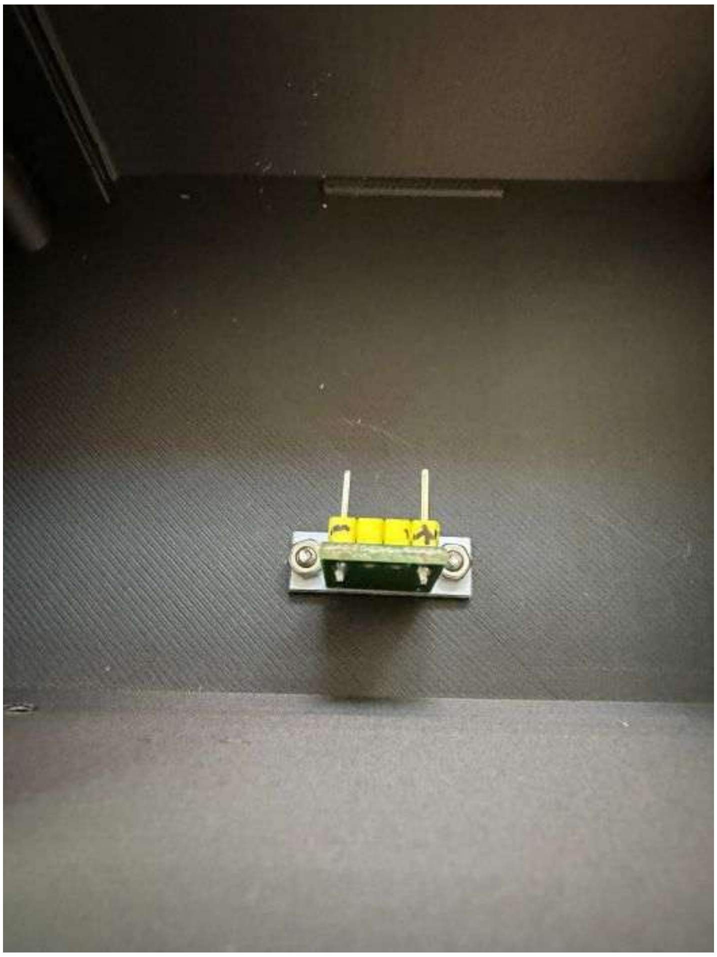
Finally, insert the tinned wire ends into the dial connector, making sure to observe the correct polarity.



8. Next, prepare the USB port for installation.
   Attach the pin header as shown in the image and solder it in place.



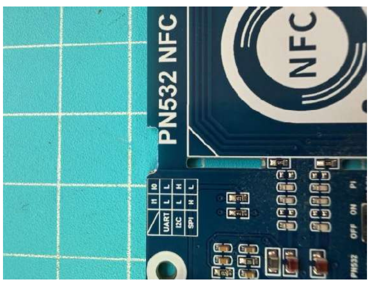
Now install the USB port into the enclosure. For this, use two M2 × 6 mm pan head screws and two nuts.



Now the display and dial can be connected. Make sure to observe the correct polarity.

9. Next, install the NFC reader.
Insert the NFC reader into the designated slot as shown in the image and connect it to the correct socket on the display.

Then hold the two housing halves together and check whether they fit properly or if a small gap appears because the display mounting screw is interfering with the NFC reader. If this is the case, a small cut-out must be made in the NFC reader.



10. Now screw the two housing halves together.
    For this, use four M2 × 8 mm hex socket screws.

    First, check whether the screws fit into the designated holes, as the printing process may have caused slight constrictions. In most cases, the four holes in the rear housing part should be checked for clearance using a 2 mm drill bit. From the opposite side, use a 5 mm drill bit to remove any slight burrs that may have formed during printing.

    Once this has been checked, the two halves can be screwed together.

11. Attach the base plate.
    First, insert the four suction cups into the designated recesses.

On the opposite side, it should look like this.

Now screw the base plate to the enclosure. For this, use M3 × 6 pan head screws with flange.
Tighten the four screws carefully so as not to overstress the threads in the housing base, which could cause damage.

That’s it – your AMS SpoolEdit is now fully assembled.
Have fun using it!